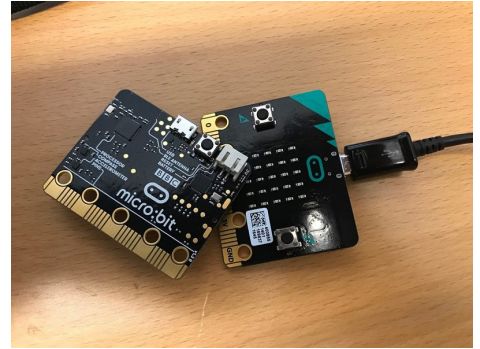


Micro:bit Kit

WHAT'S IN THE KIT

- 14 micro:bit bags per kit — each with micro:bit, battery pack, USB cable, alligator clips, jumper cables, servo motor, NeoPixel LED strip
- Demo code: Light Level Color Range, Light Level Twinkle Rate, NeoPixel Demo



QUICK-START GUIDE

1. Plug the micro:bit into a computer via USB.
2. Open makecode.microbit.org and create a project.
3. Code with blocks (or flip to Python/JavaScript for older students).
4. Click Download — drag the .hex file to the MICROBIT drive.
5. Use alligator clips to attach servos or NeoPixels for physical computing.

FULL LESSONS IN THIS GUIDE

GRADES 3-5	Name Badge + Emotion Meter	page 2
GRADES 6-8	Radio Secret Messages	page 3
GRADES 9-12	NeoPixel Data Display	page 4

PRO TIP: Pair with the Gator Science Kit for environmental science investigations.

Micro:bit Kit

LESSON · GRADES 3-5

Name Badge + Emotion Meter

LEARNING OUTCOMES

- Students will create a program with multiple inputs (buttons) and outputs (LED display)
- Students will use event blocks to trigger different behaviors
- Students will explain input → processing → output using their own project

MATERIALS

- micro:bit bags (1 per pair): board, battery, USB cable
- Chromebooks with makecode.microbit.org
- Projector for demo

INSTRUCTIONS

1. Demo the MakeCode simulator first — students can test without hardware.
2. Guided build (10 min): scrolling name badge on the LED grid.
3. Challenge (20 min): add an 'emotion meter' — button A shows a happy face, button B a calm face, shake shows surprised.
4. Pairs trade micro:bits and guess each other's button mappings.
5. Wrap-up: map the vocabulary — which parts were input, output, and program?

ACTIVITY

Name Badge + Emotion Meter — students wear their programmed badges and demo their button-triggered emotions to the class, narrating which input causes which output.

Micro:bit Kit

LESSON · GRADES 6-8

Radio Secret Messages

LEARNING OUTCOMES

- Students will send and receive data over the micro:bit radio
- Students will design and apply a simple cipher for encoding messages
- Students will evaluate the strengths and weaknesses of an encoding scheme

MATERIALS

- micro:bit bags (1 per pair)
- Chromebooks with makecode.microbit.org
- Paper for cipher keys
- Optional: radio group number cards to prevent cross-talk

INSTRUCTIONS

1. Assign each pair a radio group number so messages stay within pairs.
2. Guided build: send a number on button press, display received numbers.
3. Teams invent a cipher (e.g., 1=A shifted by 3) and write the key on paper.
4. Exchange encrypted messages by radio; partners decode using the key.
5. Spy round: a third team intercepts (same radio group) and tries to crack the code.
6. Debrief: what made codes hard to crack? Connect to real encryption and digital privacy.

ACTIVITY

Radio Secret Messages — pairs build an encrypted messaging system over micro:bit radio, then test it against classmate 'codebreakers.' The conversation about why some ciphers fell in seconds is the real lesson.

Micro:bit Kit

LESSON · GRADES 9-12

NeoPixel Data Display

LEARNING OUTCOMES

- Students will map continuous sensor data to a visual output range
- Students will program NeoPixel strips using loops and variables (or MicroPython)
- Students will justify design decisions in an ambient data display

MATERIALS

- micro:bit bags with NeoPixel strips and alligator clips
- Chromebooks with makecode.microbit.org (or python.microbit.org)
- Reference chart: sensor value ranges (light, sound, temp)

INSTRUCTIONS

1. Demo wiring: NeoPixel strip to pins with alligator clips — polarity matters.
2. Students pick a data source: sound level, light, temperature, or acceleration.
3. Build: read the sensor in a loop, map its range to LED count and color.
4. Require a written mapping spec: what value range produces what display.
5. Gallery walk with displays running; visitors must interpret each display cold.
6. Discussion: what makes an ambient display readable without explanation?

ACTIVITY

NeoPixel Data Display — students build an ambient visualization (a 'noise thermometer' for the classroom, a light meter, a motion indicator) and test whether strangers can read it without instructions — a real data-visualization design constraint.